University of Technical Education HCMC
Faculty of Information Technology
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SOFTWARE DEVELOPMENT ENVIRONMENTS

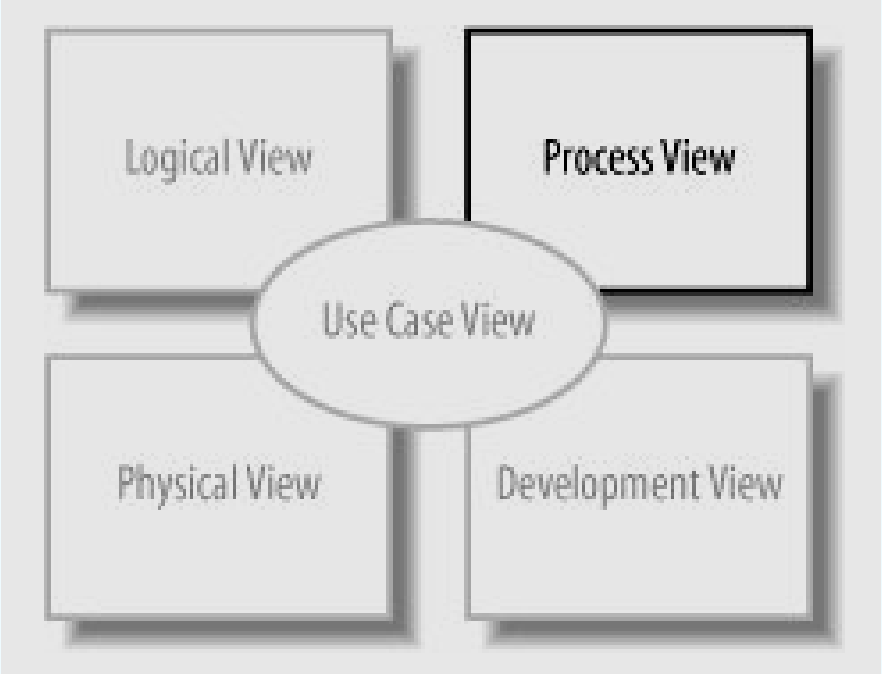
Chapter 02

Unified Modeling Language - UML

Lecturer: Nguyễn Trần Thi Văn

Lesson 05

Activity Diagram



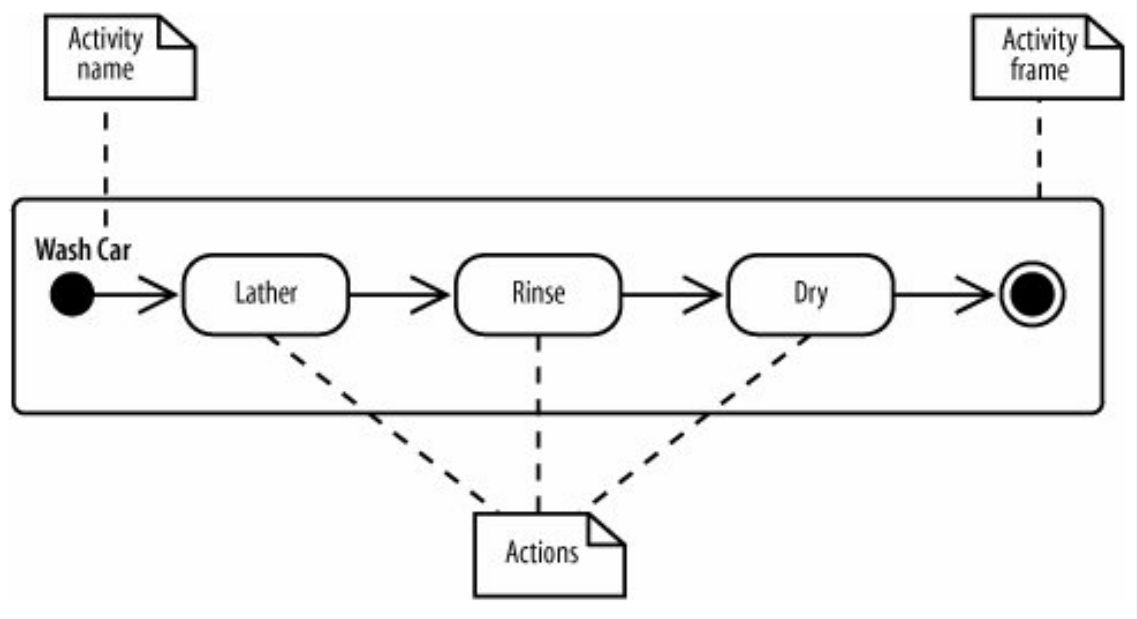
What is an Activity Diagram?

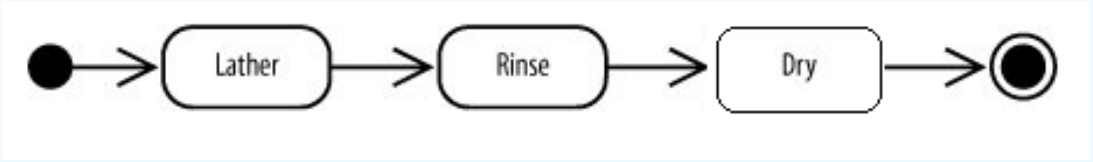
- Shows how the system completes the goals in the use case diagram
- Representing processes occurring in the system by using chains of actions
- Usually used to model business processes.
- Is the only diagram of the process view in UML



Activities and actions

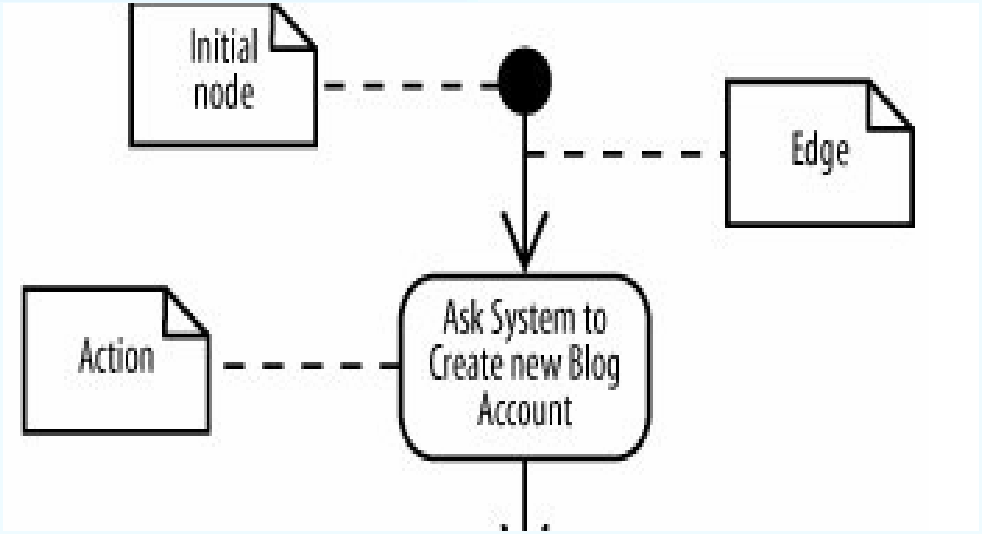
- An activity consists of many actions
- Activity name and activity frame may be omited





Components

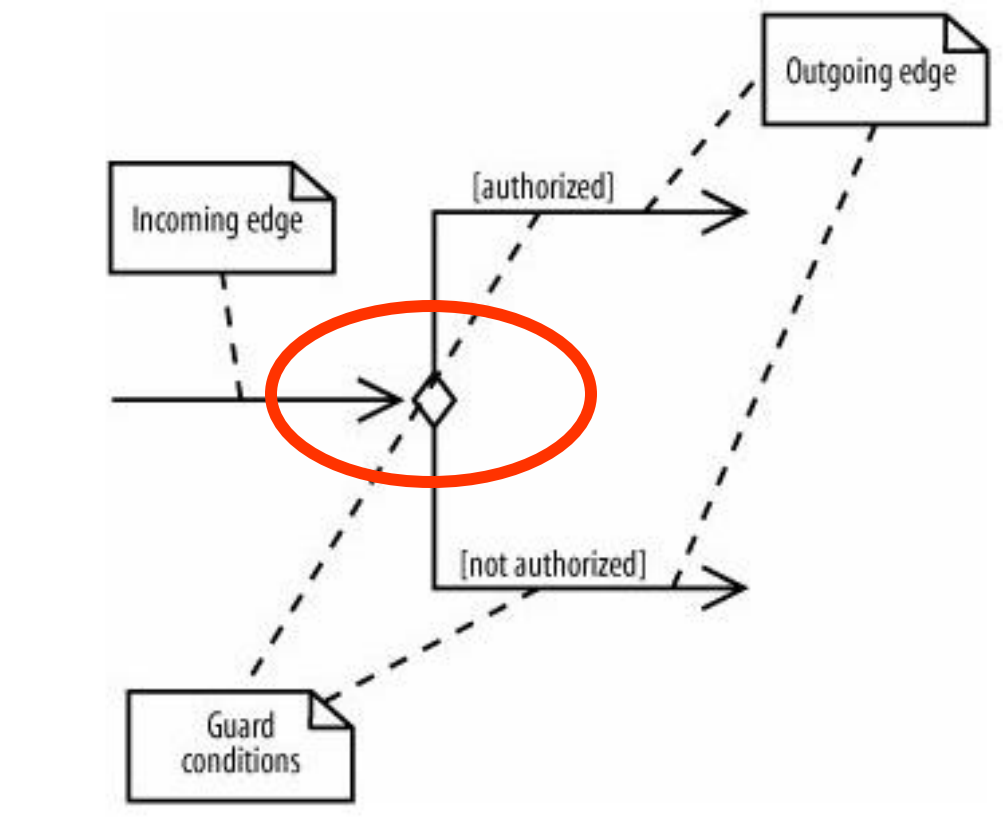
- Initial node
- Edge
- Action
- Decision node
- Merge
- Folk & join
- Time event
- Connectors
- Final node





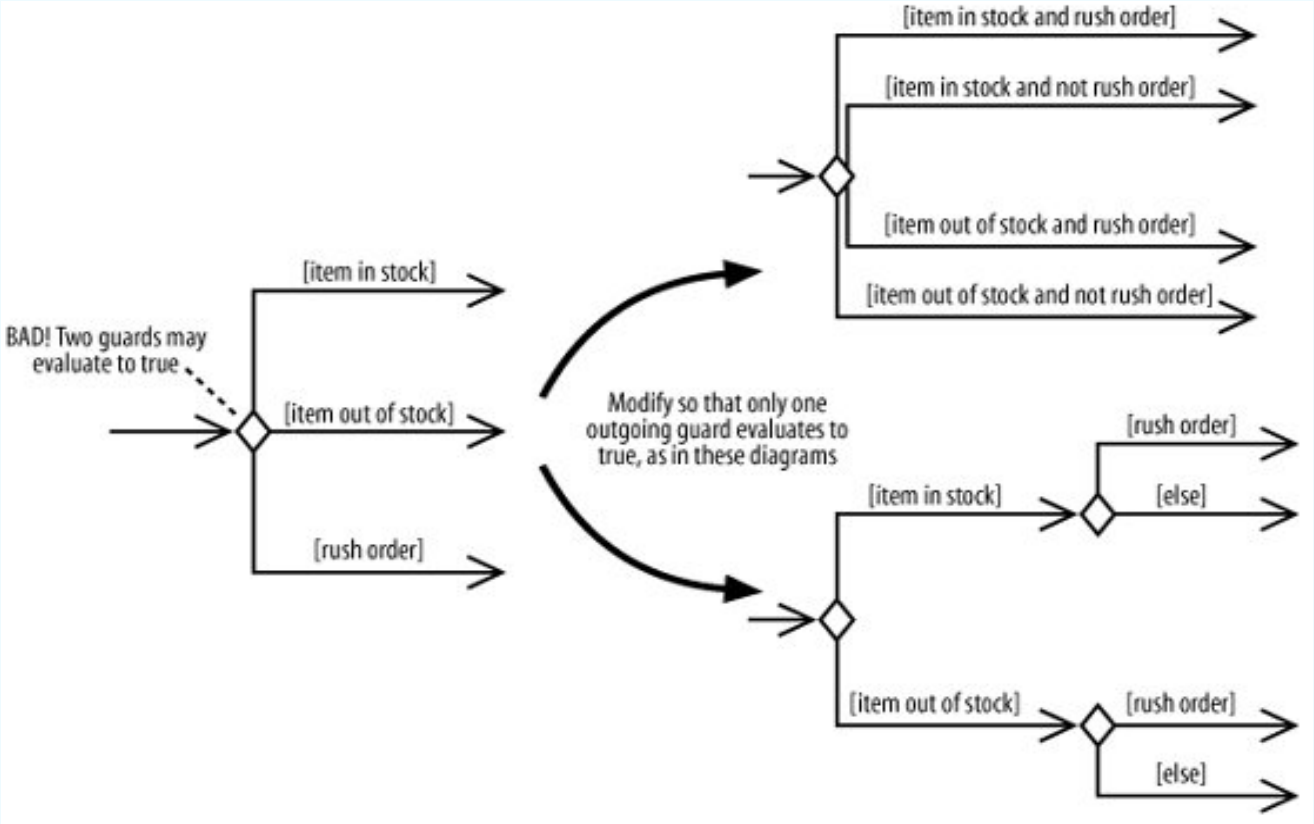
Decision node

- 1 incoming edge with 2 or more outgoing edges, each with 1 guard conditions



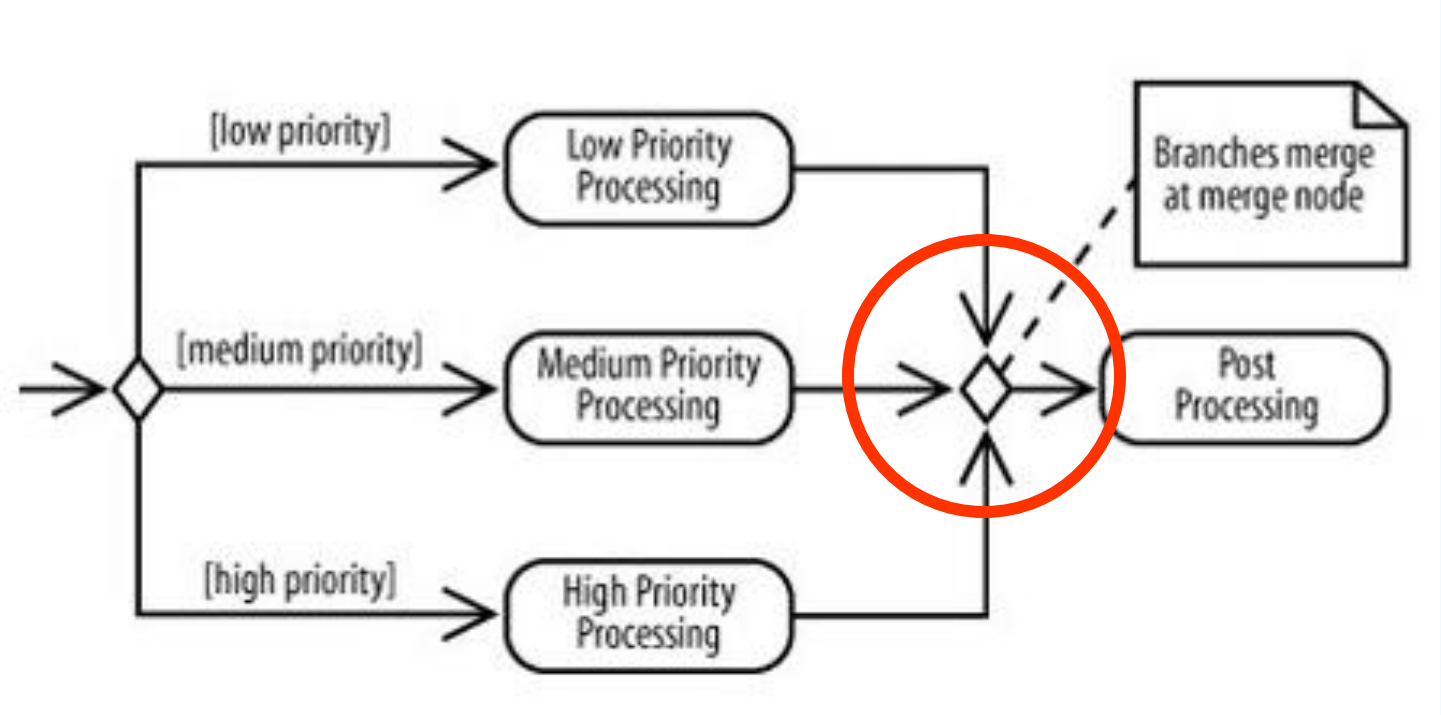
Decision node (2)

- **Any 2 guard conditions of the same node must not be true at the same time**



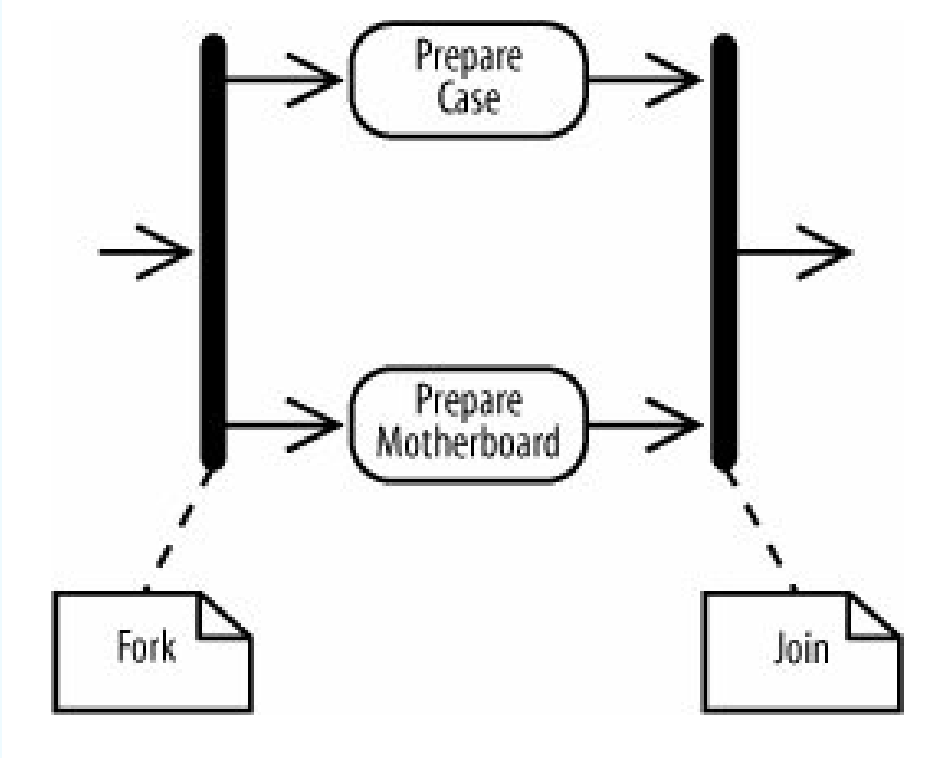
Merge node

- **Used to merge multiple edges before processing the next action**



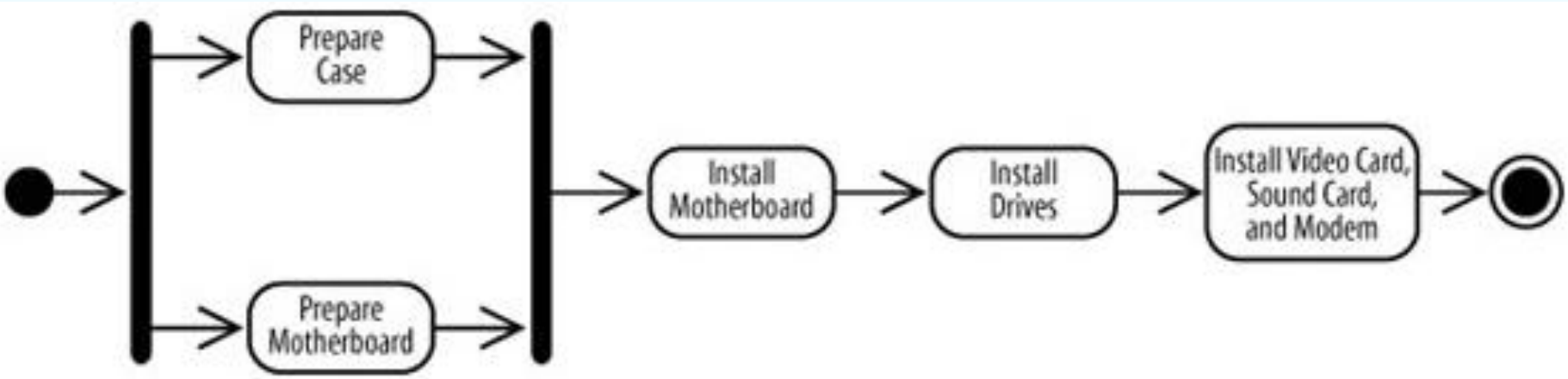
Folks and joins

- **Used to split and join actions that can be processed in parallel at the same time**



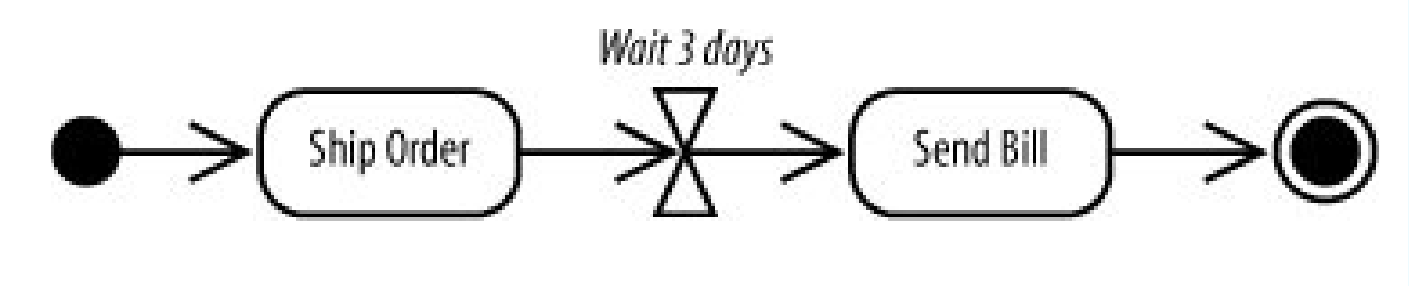
Folks and joins – an example

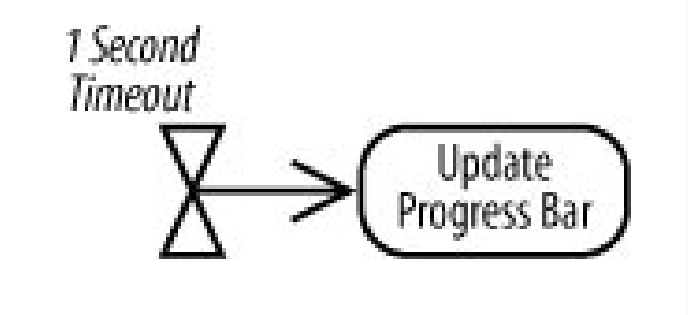
- **The computer assembly workflow**



Time events

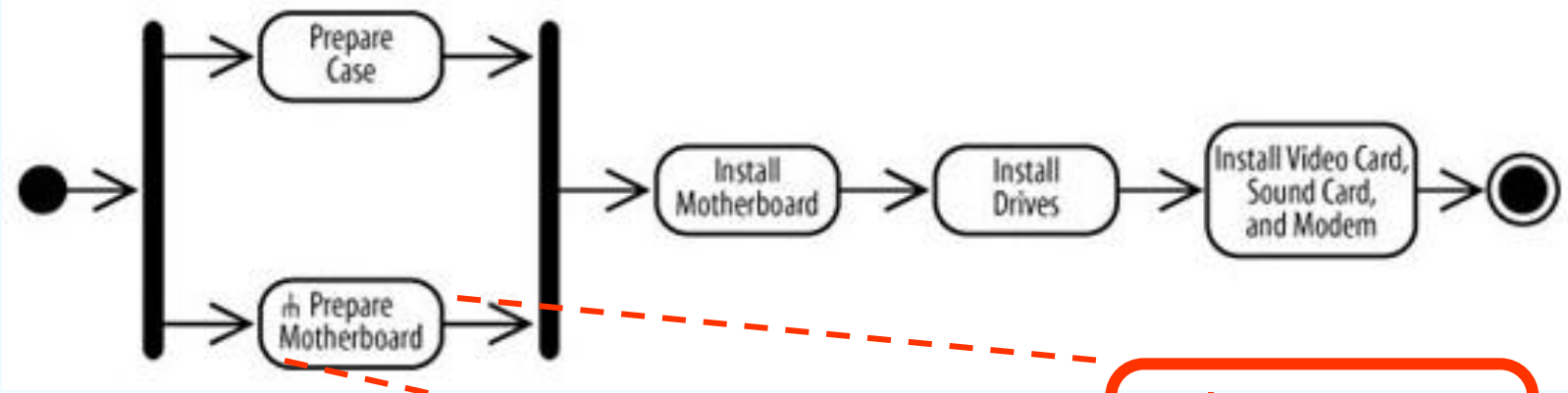
- Represented by an hour-glass with the period above
- Used to model a wait period
- A time event with no incoming flow → repeated many times

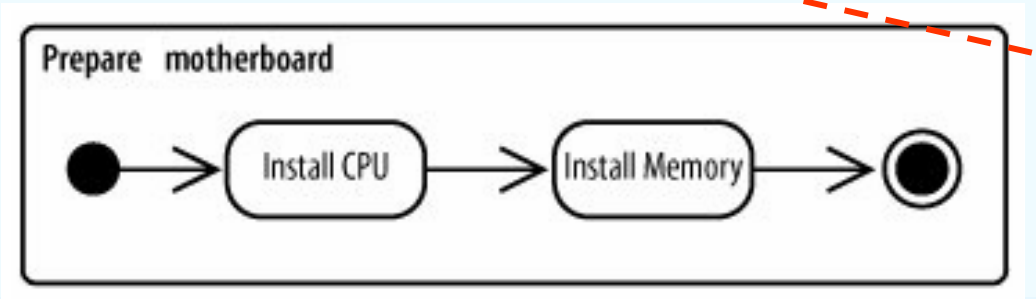




Calling other activities

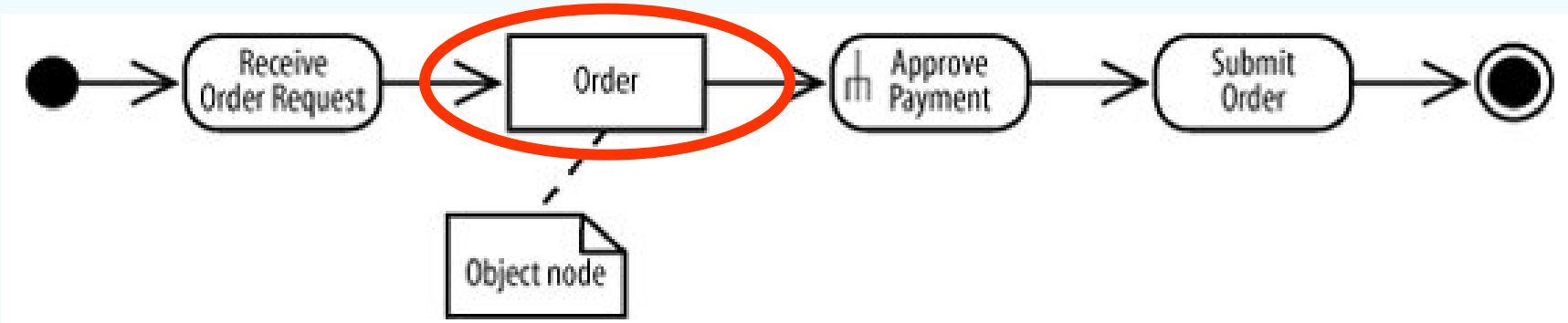
- Use the folk-down symbol before the action name
- Used to reduce complexity in one diagram





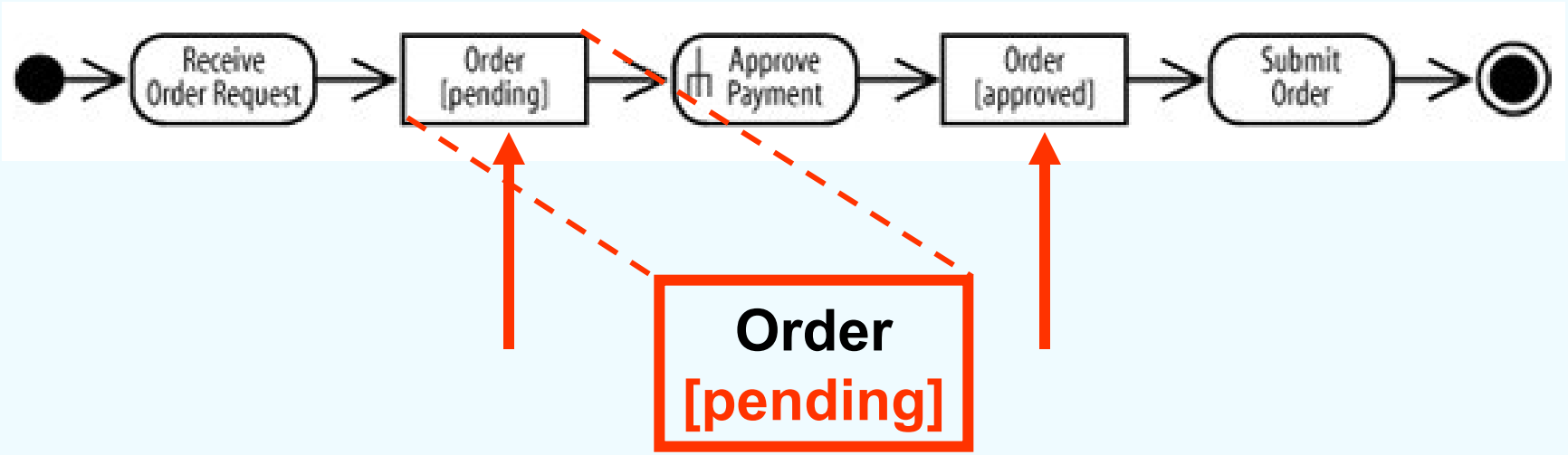
Object passing

- Emphasize the important data in the activity
- Show which action relating to that data



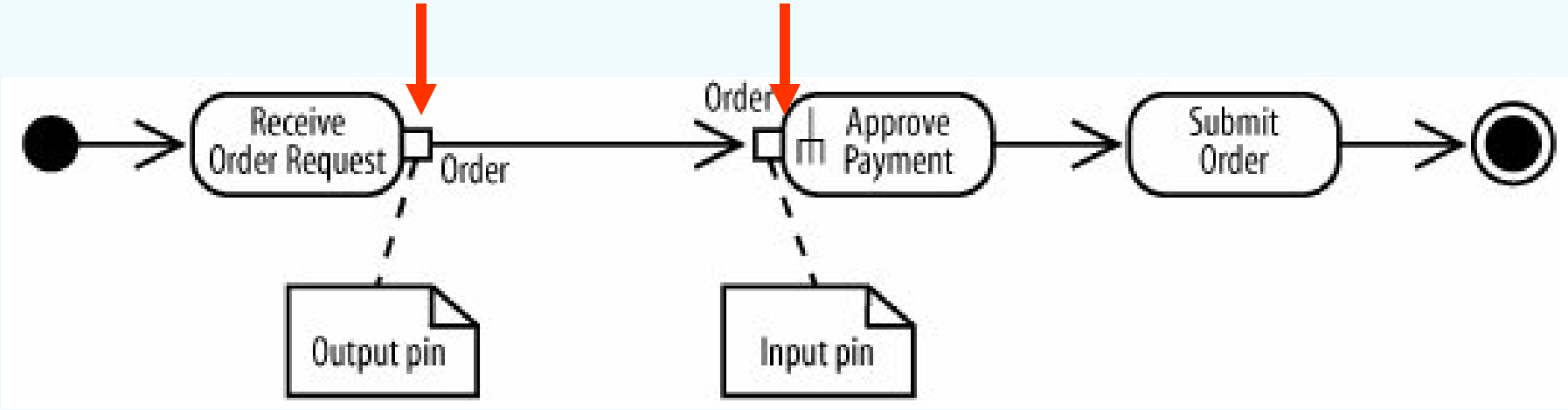
Object state transition

- **The states are shown in brackets**



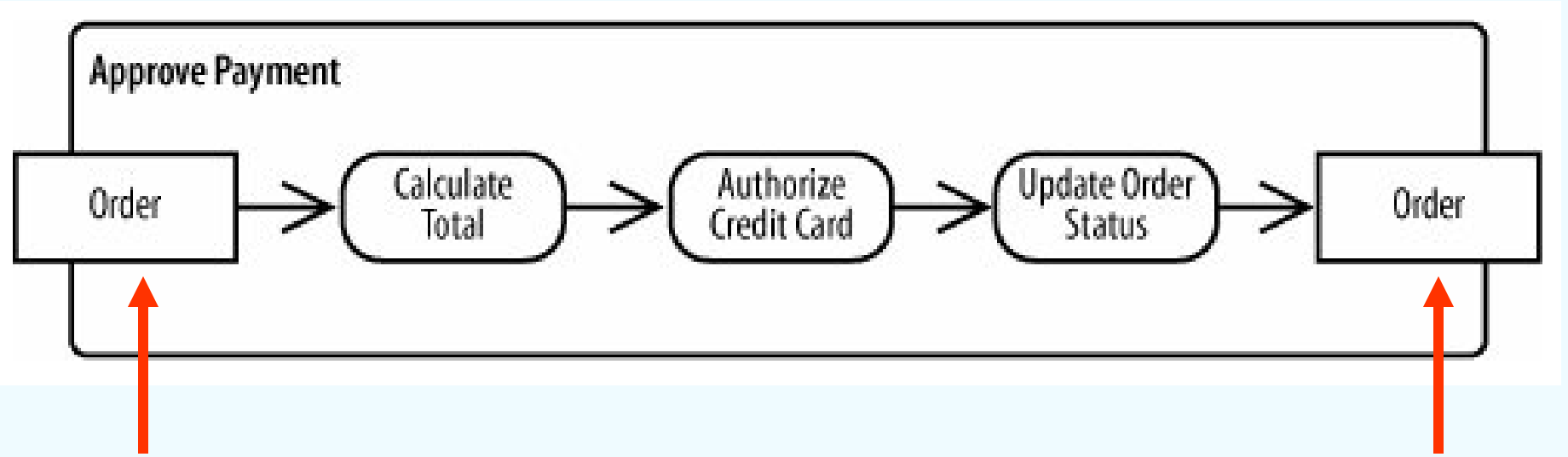
Action inputs and outputs

- **These pins are used when 1 action passes one or more object(s) to its subsequent action**



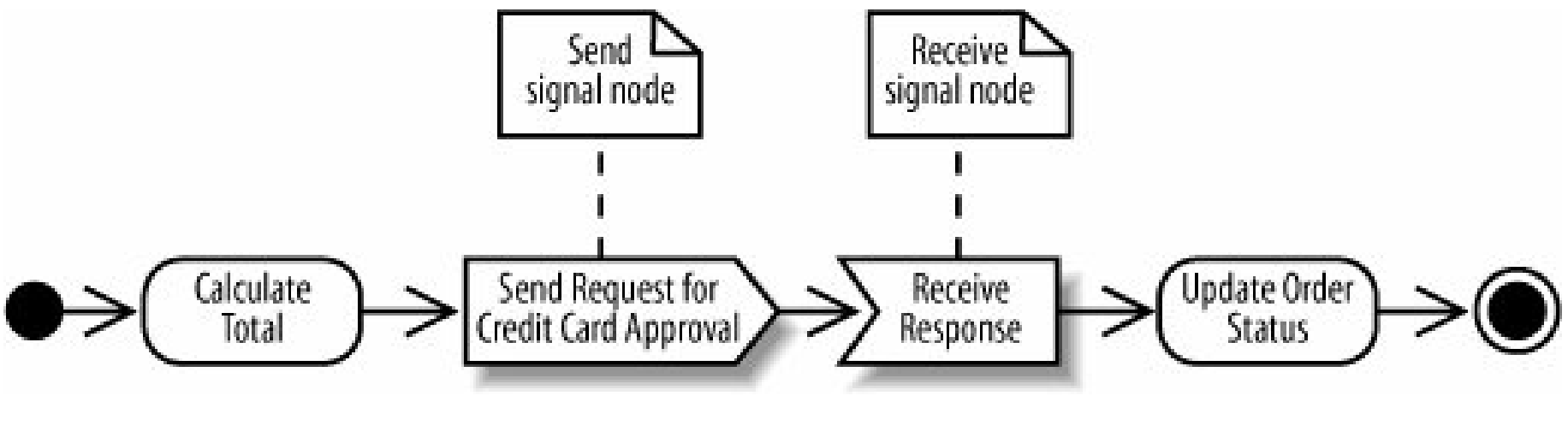
Activity inputs and outputs

- **Object nodes can be used to emphasize the input and output of an activity**



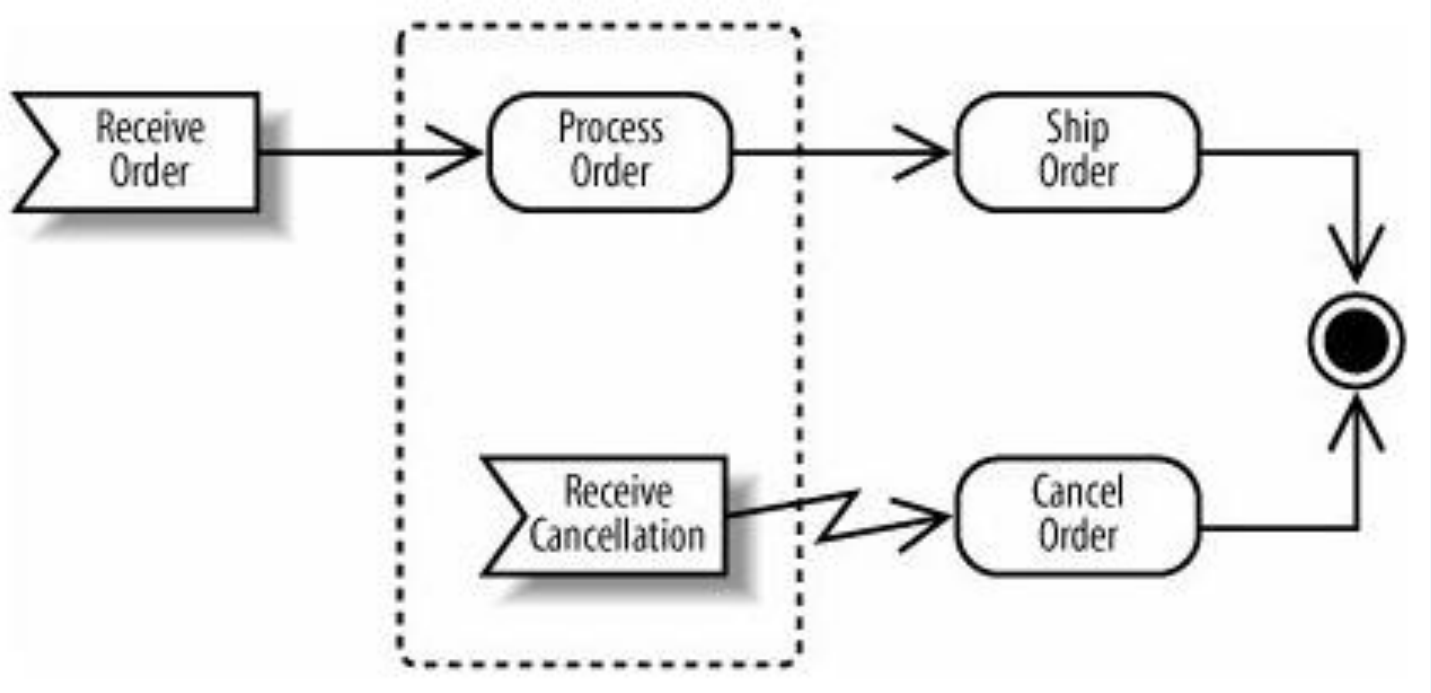
Sending and receiving signals

- **Used to express that the process communicates with external participants**



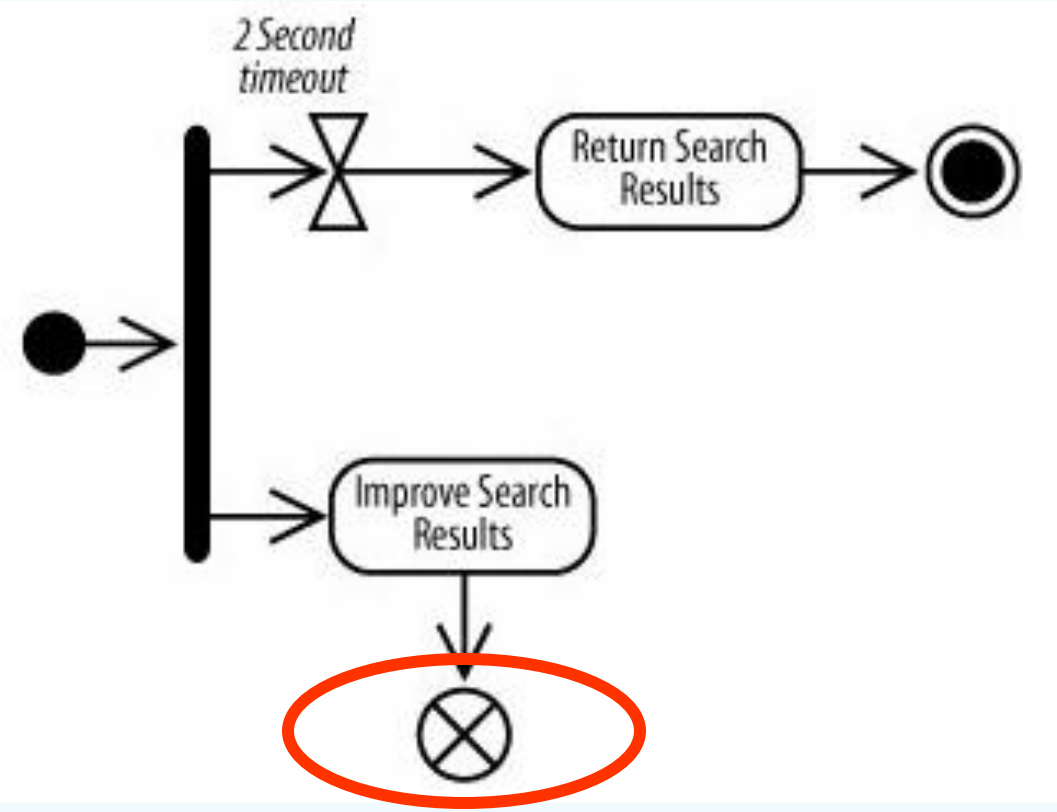
Interrupting an activity

- **Interruption region shows that the process can be interrupted during the period**

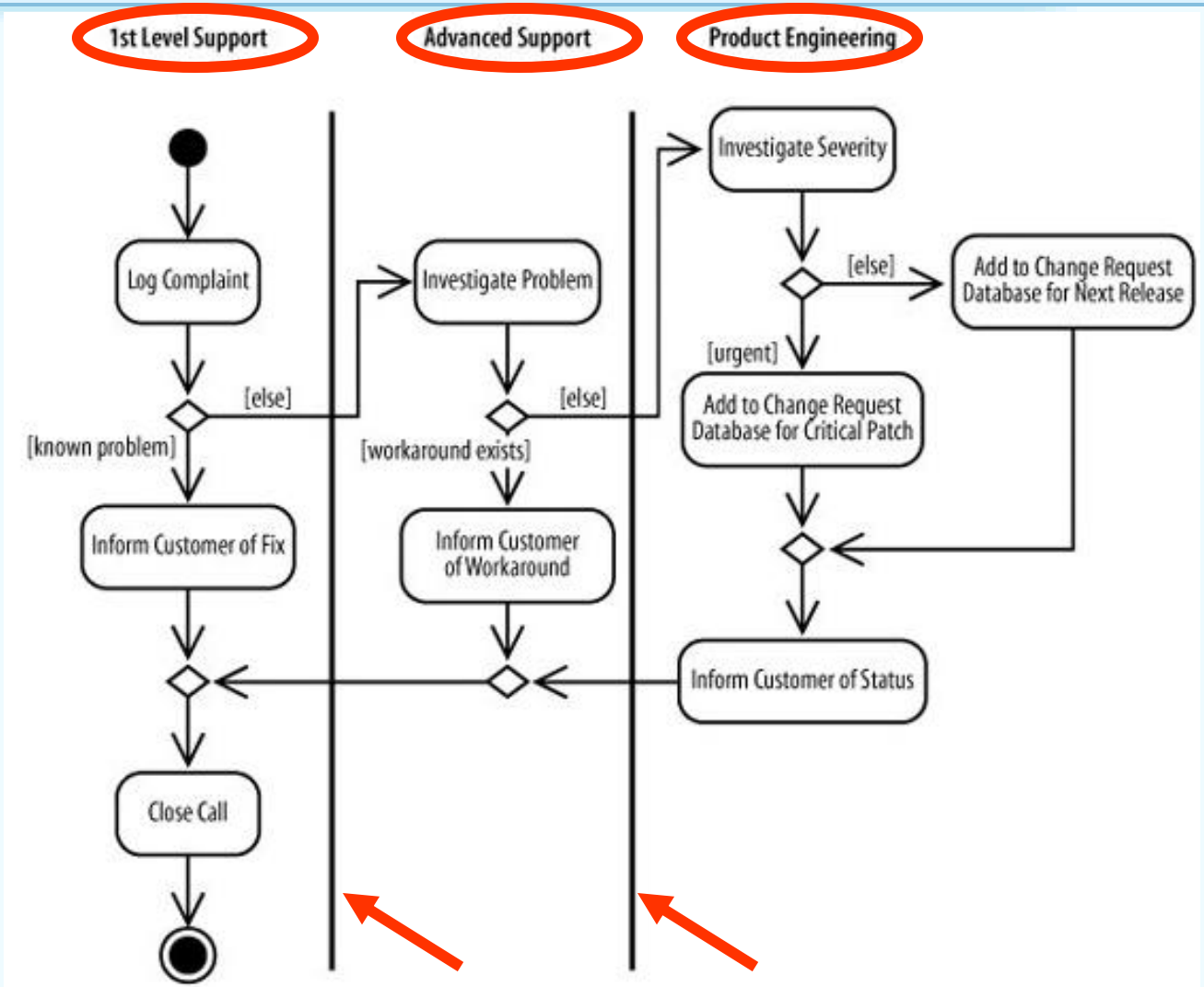


Ending a flow

- The symbol is used to mark the ending of a flow, not the whole activity

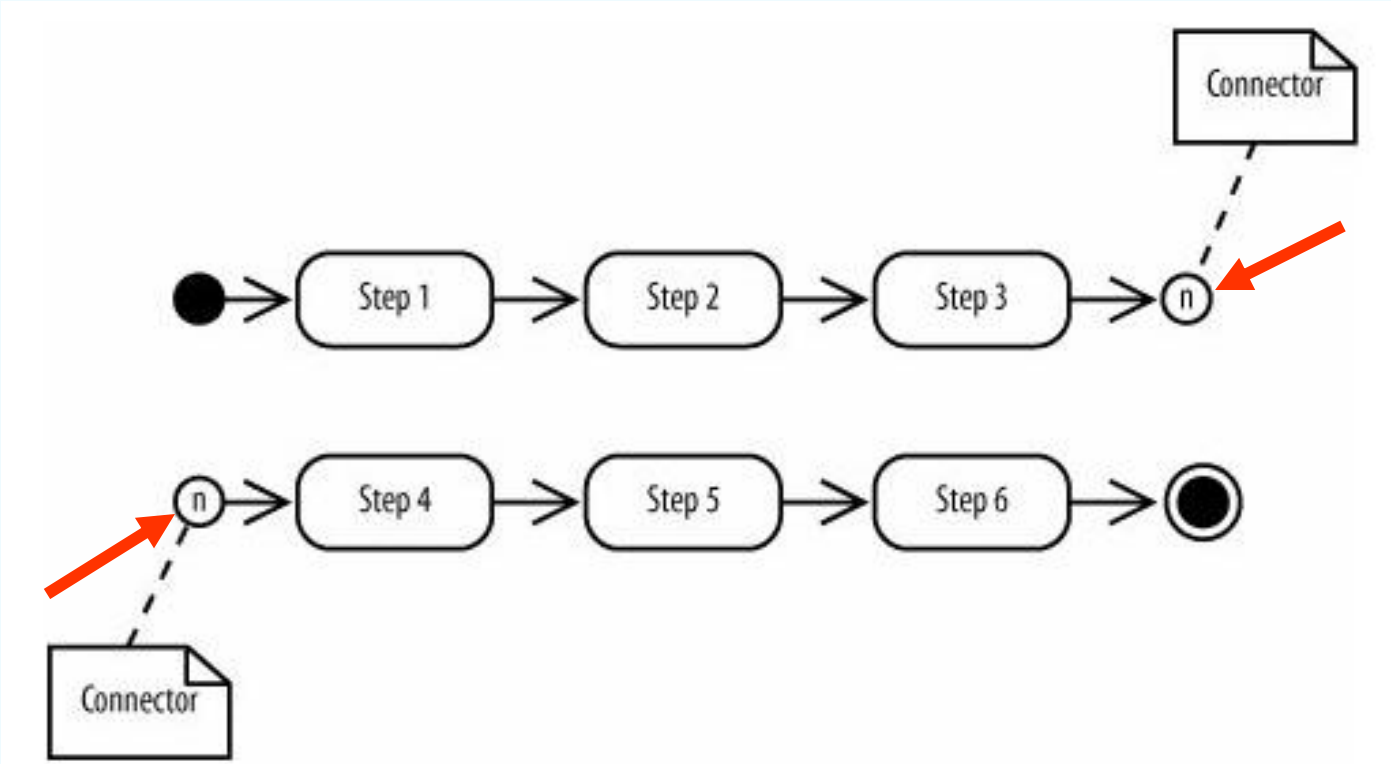


Partitions (swimlanes)



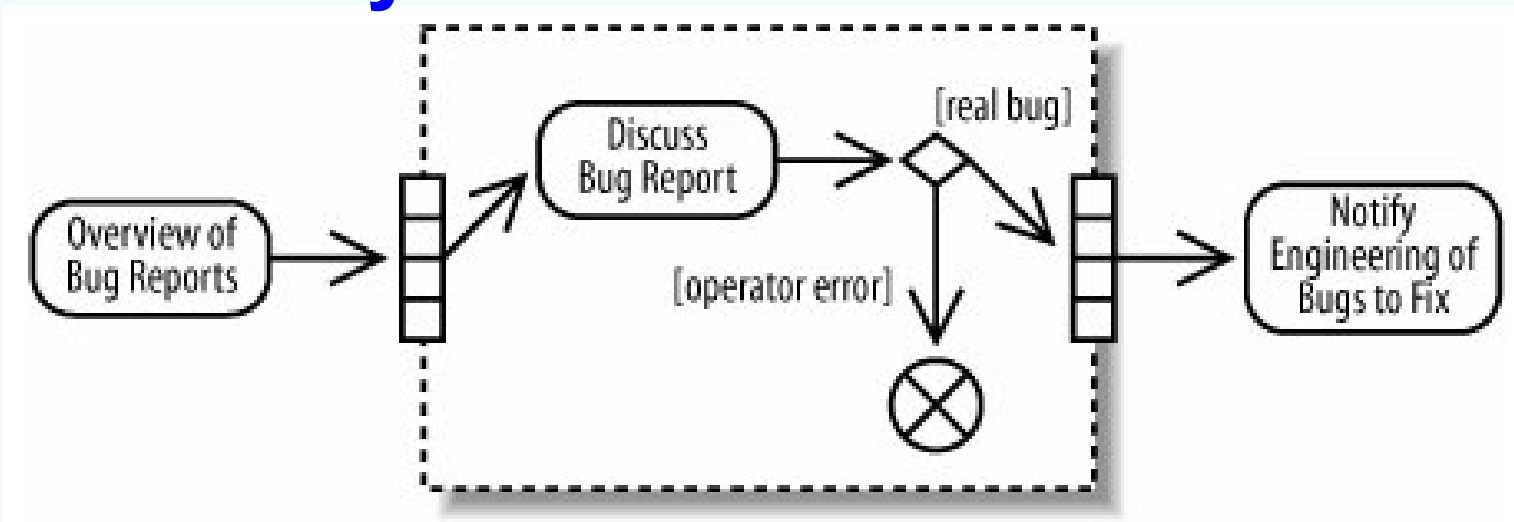
Connectors

- **Used to manage the complexity of the activity diagram (by fragmentation)**



Expansion regions

- Covered in a dashed rounded rectangle with 4 aligned boxes at either side – Input and output are collections
- The actions in the region are repeated with every element of the collection





THE END

